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Register N.O: 22BCI0052

Course: Malware Analysis Lab

Digital Assessment:1

Q1.

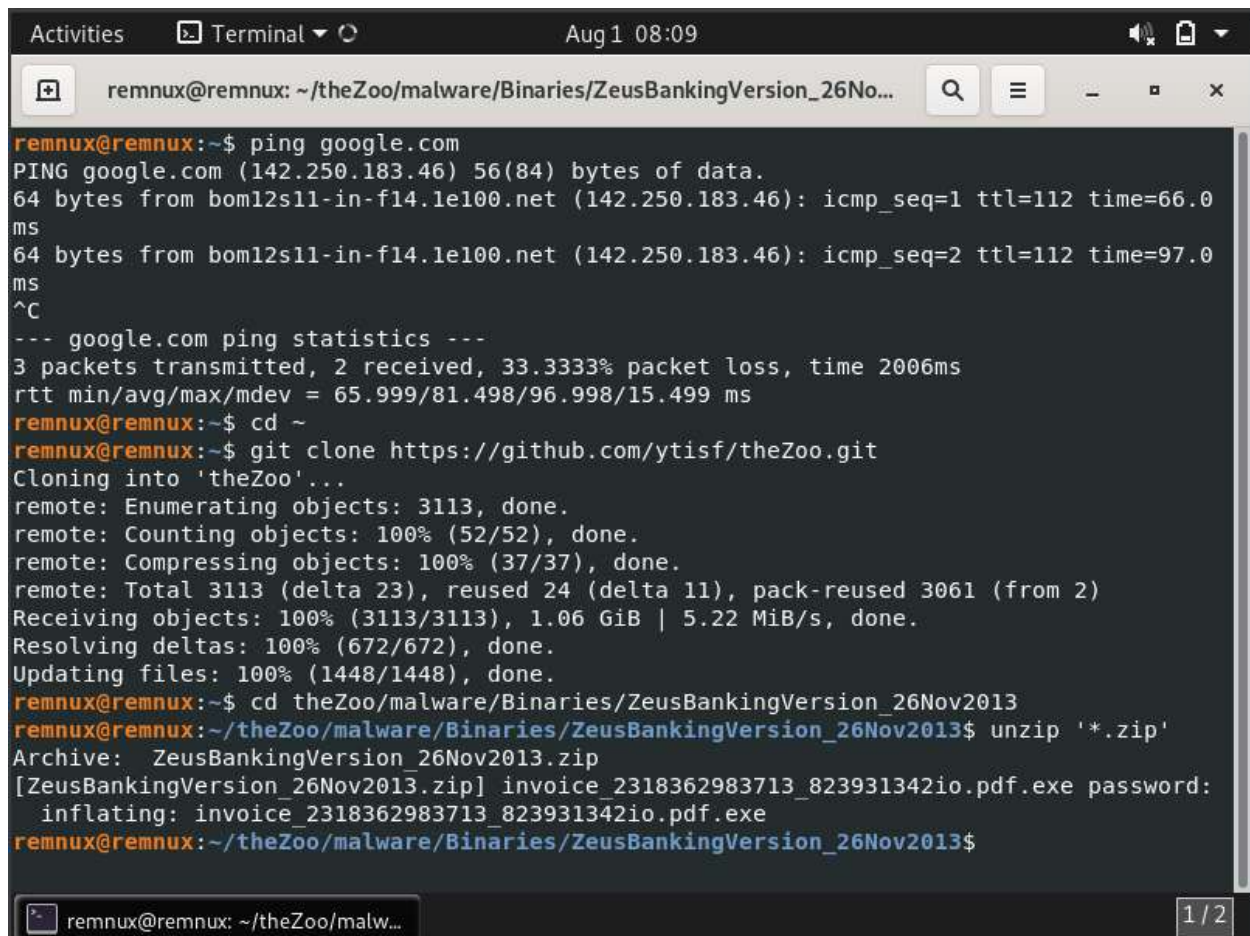
Aim: To analyze the Zeus Banking Trojan using VirtualBox by configuring REMnux as the attacker (analysis environment) and Windows 10 as the victim (execution environment), performing both static and dynamic malware analysis.

Tools Used:

1. VirusTotal
2. PeStudio
3. FLOSS
4. Capa
5. Cutter
6. INetSim
7. Process Monitor
8. Wireshark

Procedure:

Extracting the zeus banking trojan inside the remnux using git and unzipping the zip file

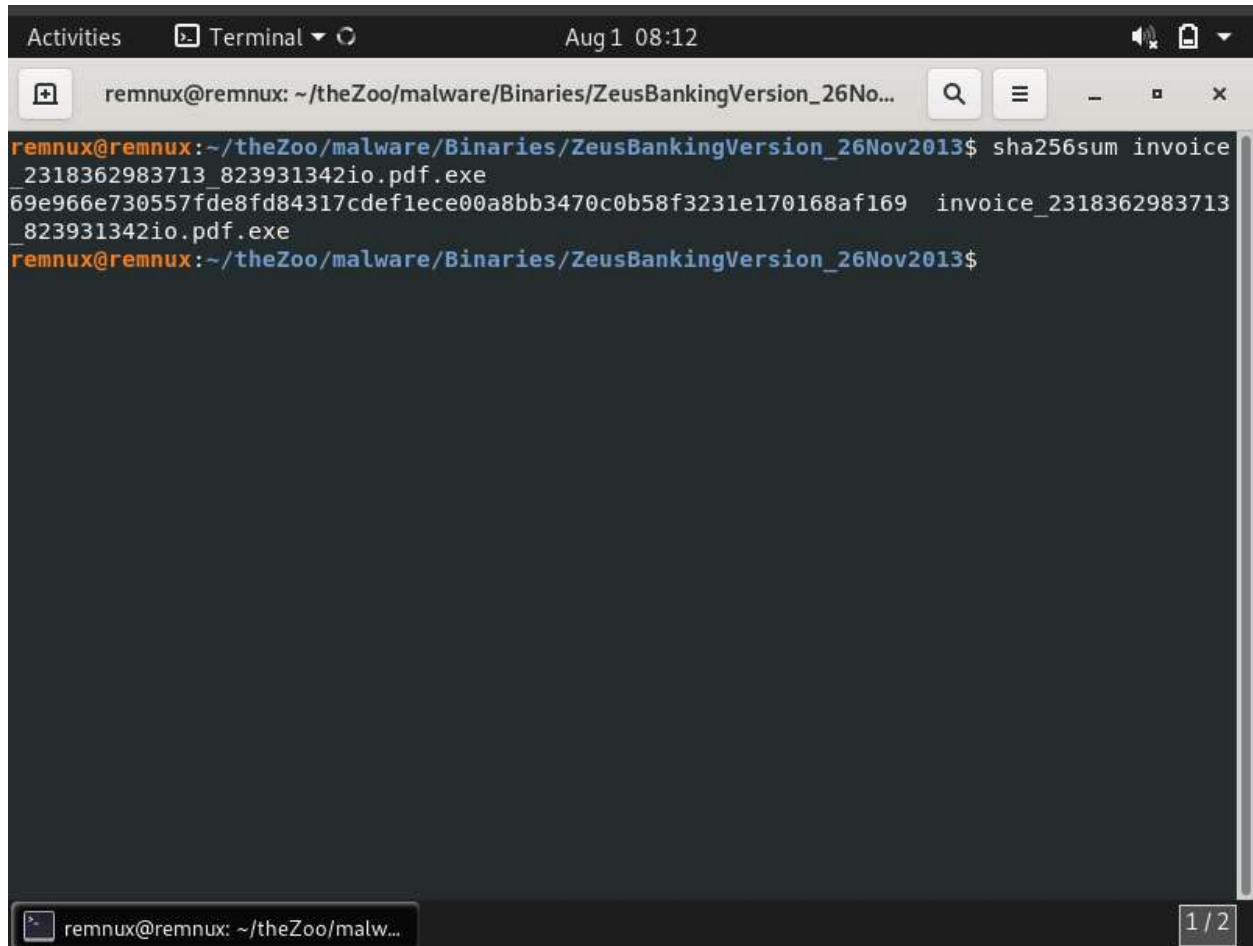


```
Activities  Terminal  Aug 1 08:09
remnux@remnux: ~/theZoo/malware/Binaries/ZeusBankingVersion_26No...

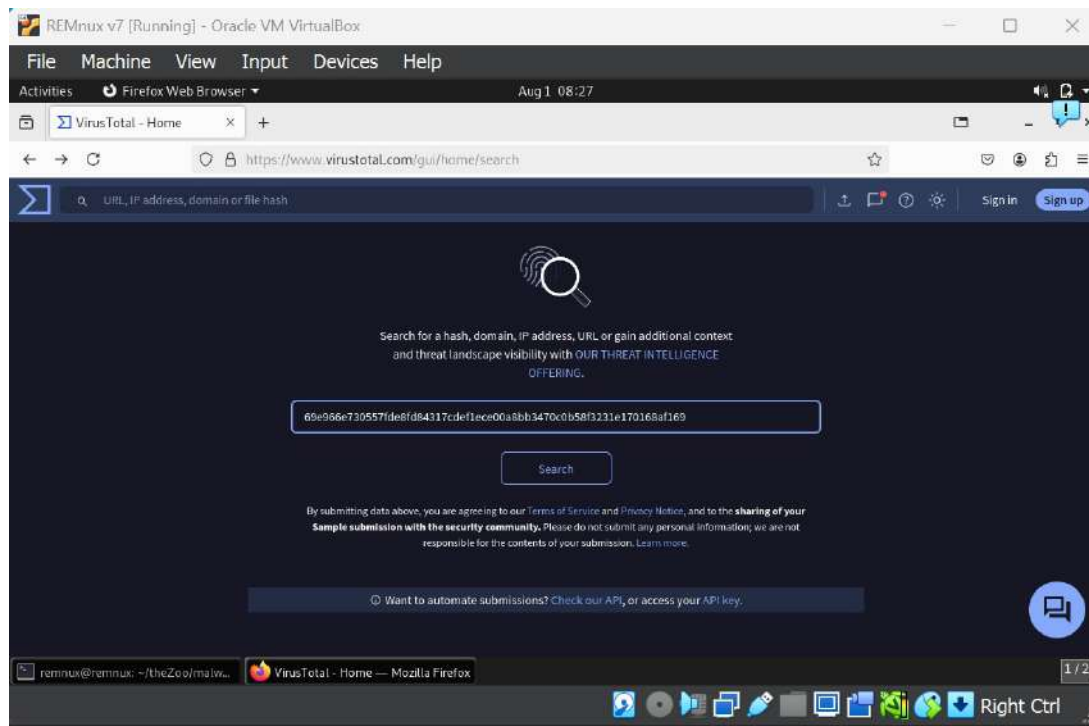
remnux@remnux:~$ ping google.com
PING google.com (142.250.183.46) 56(84) bytes of data.
64 bytes from bom12s11-in-f14.1e100.net (142.250.183.46): icmp_seq=1 ttl=112 time=66.0 ms
64 bytes from bom12s11-in-f14.1e100.net (142.250.183.46): icmp_seq=2 ttl=112 time=97.0 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 2 received, 33.333% packet loss, time 2006ms
rtt min/avg/max/mdev = 65.999/81.498/96.998/15.499 ms
remnux@remnux:~$ cd ~
remnux@remnux:~$ git clone https://github.com/ytisf/theZoo.git
Cloning into 'theZoo'...
remote: Enumerating objects: 3113, done.
remote: Counting objects: 100% (52/52), done.
remote: Compressing objects: 100% (37/37), done.
remote: Total 3113 (delta 23), reused 24 (delta 11), pack-reused 3061 (from 2)
Receiving objects: 100% (3113/3113), 1.06 GiB | 5.22 MiB/s, done.
Resolving deltas: 100% (672/672), done.
Updating files: 100% (1448/1448), done.
remnux@remnux:~$ cd theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013
remnux@remnux:~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013$ unzip '*.zip'
Archive:  ZeusBankingVersion_26Nov2013.zip
[ZeusBankingVersion_26Nov2013.zip] invoice_2318362983713_823931342io.pdf.exe password:
  inflating: invoice_2318362983713_823931342io.pdf.exe
remnux@remnux:~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013$
```

1. VirusTotal

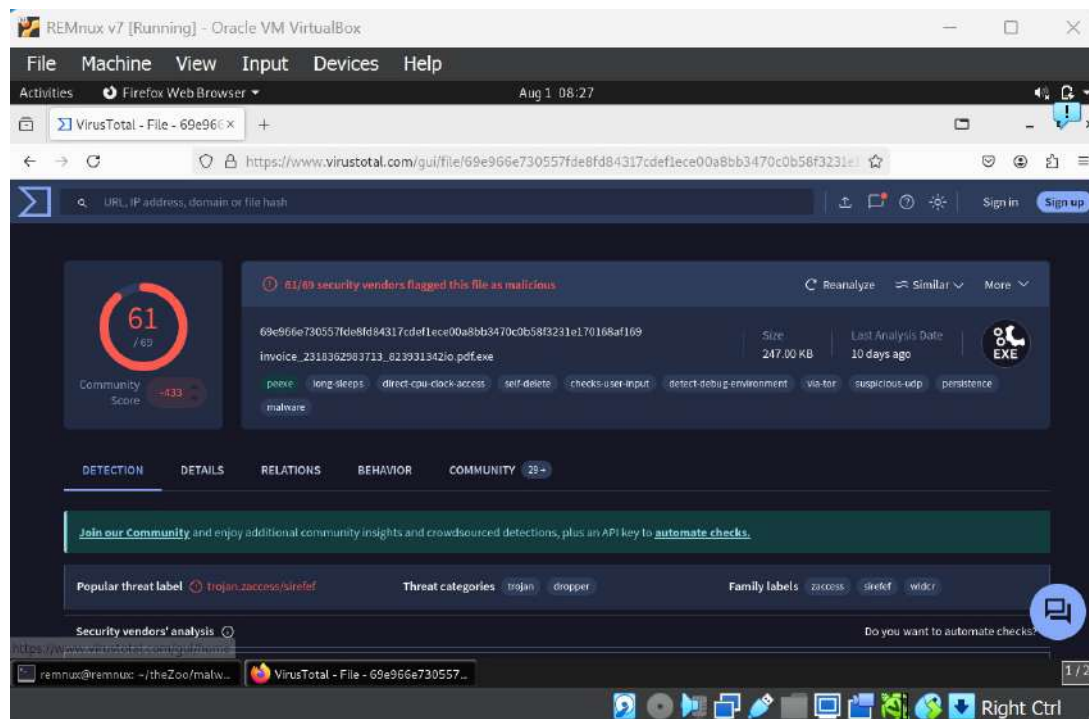
Finding sha256 value

A screenshot of a Linux terminal window. The window title bar shows 'Activities', 'Terminal', and the time 'Aug 1 08:12'. The terminal content shows a user at the 'remnux' prompt in the directory '~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013'. The user has executed the command 'sha256sum invoice_2318362983713_823931342io.pdf.exe'. The output is a long hexadecimal string followed by the filename. The terminal window has a dark background and a light-colored title bar.

```
remnux@remnux: ~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013$ sha256sum invoice_2318362983713_823931342io.pdf.exe
69e966e730557fde8fd84317cdef1ece00a8bb3470c0b58f3231e170168af169  invoice_2318362983713_823931342io.pdf.exe
remnux@remnux: ~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013$
```

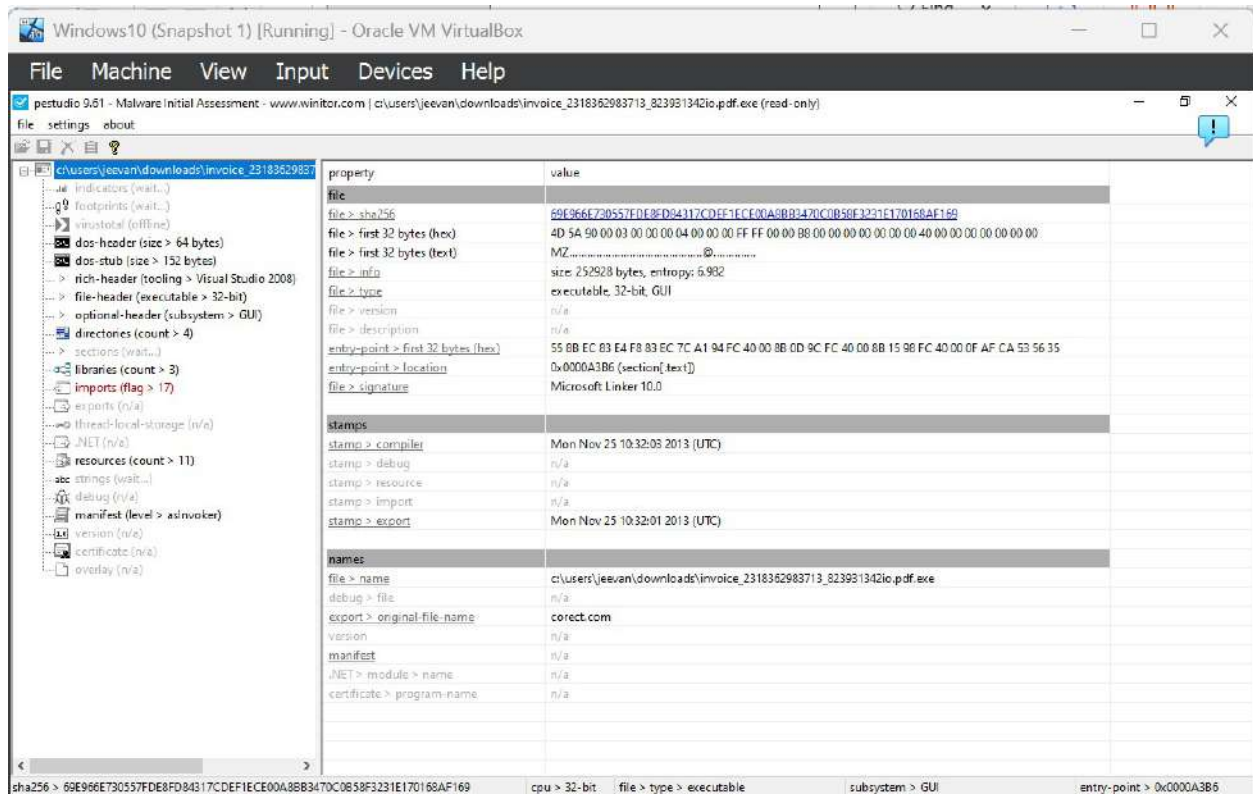


VirusTotal result:



2. PeStudio

Shared the zeus trojan folder from remnux to windows
10 vm via python



PeStudio Result:

Windows10 (Snapshot 1) [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\jeevan\downloads\invoice_2318362983713_823931342io.pdf.exe (read-only)

file settings about

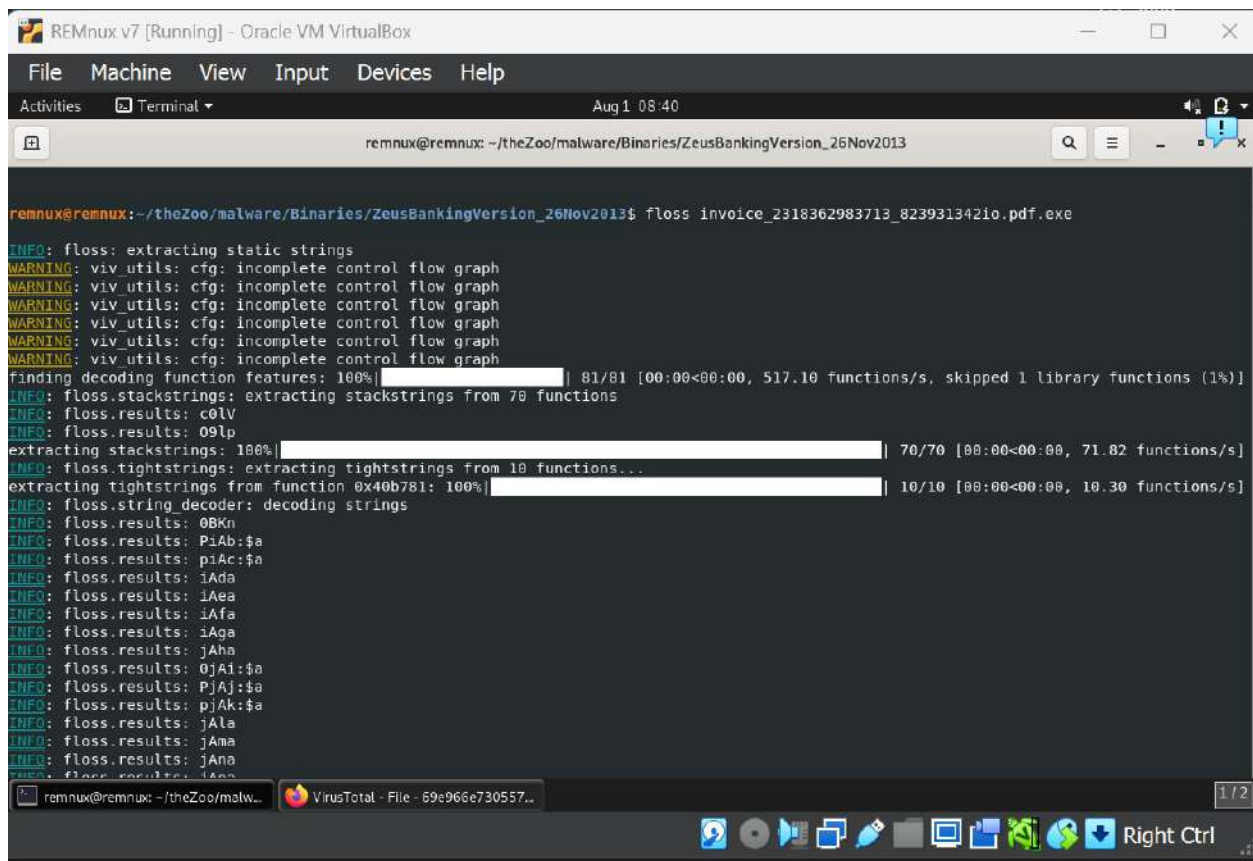
c:\users\jeevan\downloads\invoice_2318362983713

- indicators (sections > self-modifying)
- footprints (type > sha256)
- virustotal (offline)
- dos-header (size > 64 bytes)
- dos-stub (size > 152 bytes)
- rich-header (tooling > Visual Studio 2008)
- file-header (executable > 32-bit)
- optional-header (subsystem > GUI)
- directories (count > 4)
- sections (characteristics > self-modifying)
- libraries (count > 3)
- imports (flag > 17)
- exports (n/a)
- thread-local storage (n/a)
- NET (n/a)
- resources (count > 11)
- strings (count > 1416)
- debug (n/a)
- manifest (level > asInvoker)
- version (n/a)
- certificate (n/a)
- overlay (n/a)

indicator (19)	detail
file > name	c:\users\jeevan\downloads\invoice_2318362983713_823931342io.pdf.exe
file > signature	Microsoft Linker 10.0 Visual Studio 2008
file > sha256	69E966E730557FD8FD84317CDEF1ECE00A8B83470C0B56F3231E170166A...
file > info	size: 252928 bytes, entropy: 6.982
file > type	executable, 32-bit, GUI
virustotal > score	The server name or address could not be resolved
stamp > computer	Mon Nov 25 10:32:03 2013
languages > names	English-US unknown
resources > info	count: 11, size: 22178 bytes, file-ratio: 8.77%
manifest > general	name: n/a, description: n/a, severity: asInvoker
file > version	n/a
entry-point > location	0x0000A3B5 (section: .text)
sections > self-modifying	name: .pdata
certificate	n/a
imports > ordinal > count	3
imports > flag	AllowSetForegroundWindow DdeQueryNextServer EnumClipboardFo...
imphash > md5	459E320C8A8F7E7049FF769CD123FD31
exports	n/a
overlay	n/a

sha256 > 69E966E730557FD8FD84317CDEF1ECE00A8B83470C0B56F3231E170166AF169 cpu > 32-bit file > type > executable subsystem > GUI entry-point > 0x0000A3B5

3. FLOSS



The screenshot shows a terminal window within a REMnux v7 virtual machine. The user has executed the command `floss invoice_2318362983713_82393134210.pdf.exe`. The output displays various analysis steps and results, including static string extraction, control flow graph warnings, stackstring extraction, and a list of decoded strings.

```
remnux@remnux: ~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013$ floss invoice_2318362983713_82393134210.pdf.exe
INFO: floss: extracting static strings
WARNING: viv_utils: cfg: incomplete control flow graph
WARNING: viv_utils: cfg: incomplete control flow graph
WARNING: viv_utils: cfg: incomplete control flow graph
WARNING: viv_utils: cfg: incomplete control flow graph
WARNING: viv_utils: cfg: incomplete control flow graph
WARNING: viv_utils: cfg: incomplete control flow graph
finding decoding function features: 100% [REDACTED] | 81/81 [00:00<00:00, 517.10 functions/s, skipped 1 library functions (1%)]
INFO: floss.stackstrings: extracting stackstrings from 78 functions
INFO: floss.results: c0lv
INFO: floss.results: 09lp
extracting stackstrings: 100% [REDACTED] | 70/70 [00:00<00:00, 71.82 functions/s]
INFO: floss.tightstrings: extracting tightstrings from 10 functions...
extracting tightstrings from function 0x40b781: 100% [REDACTED] | 10/10 [00:00<00:00, 10.30 functions/s]
INFO: floss.string decoder: decoding strings
INFO: floss.results: 0BKn
INFO: floss.results: PiAb:$a
INFO: floss.results: piAc:$a
INFO: floss.results: iAda
INFO: floss.results: iAea
INFO: floss.results: iAfa
INFO: floss.results: iAga
INFO: floss.results: jAha
INFO: floss.results: 0jA1:$a
INFO: floss.results: PjAJ:$a
INFO: floss.results: pjAk:$a
INFO: floss.results: jAla
INFO: floss.results: jAma
INFO: floss.results: jAna
INFO: floss.results: iAaa
```

Floss Result:

REMnux v7 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Aug 1 08:48

remnux@remnux: ~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013

```
INFO: floss.results: pmA2:$a$
INFO: floss.results: mA{:$a$
INFO: floss.results: mA|:$a$
INFO: floss.results: mA}:$a$
INFO: floss.results: mA~:$a$
INFO: floss.results: 08Xu
INFO: floss.results: 09lp
emulating function 0x40baf9 (call 1/1): 100% | 20/20 [00:06-00:00, 3.31 functions/s]
INFO: floss: finished execution after 70.19 seconds
INFO: floss: rendering results

FLARE FLOSS RESULTS (version v3.1.0-0-gdb9af41)

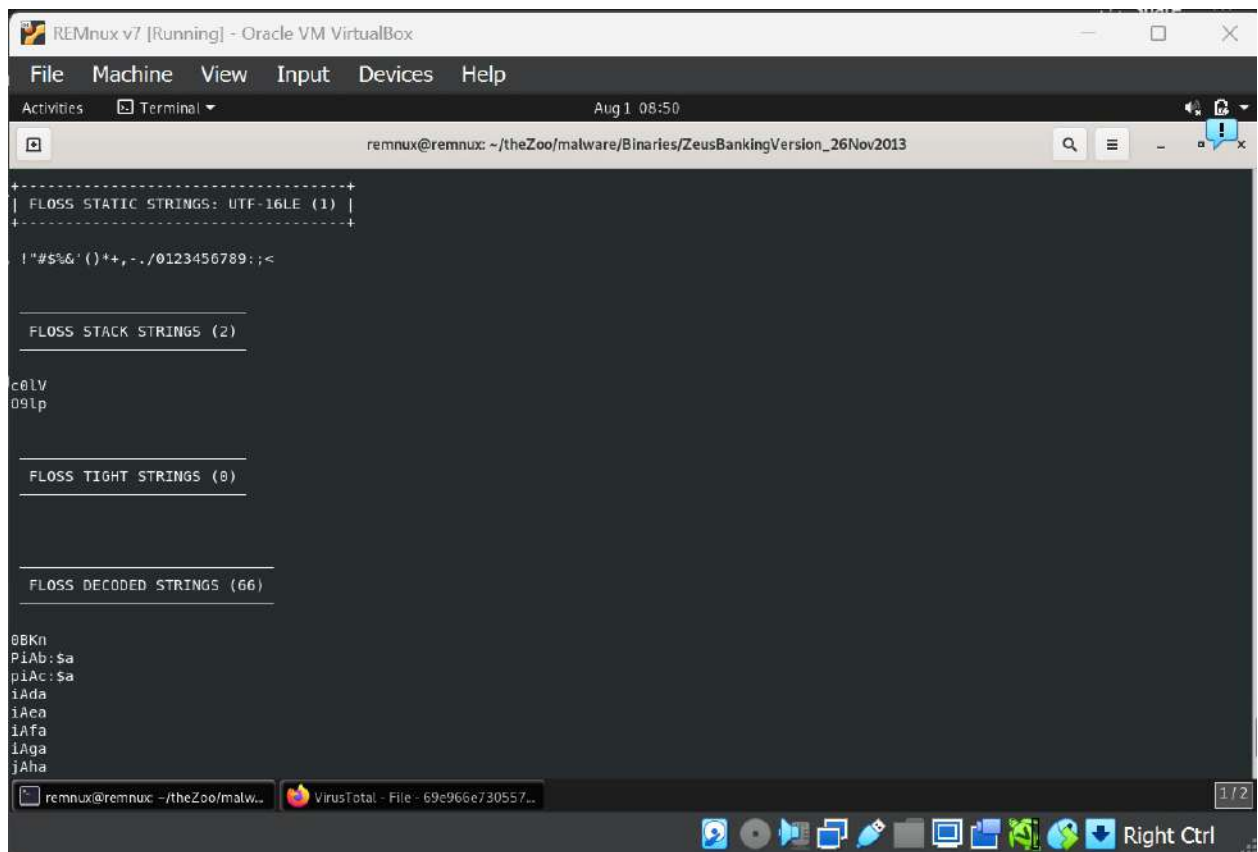
+-----+
| file path          | invoice_2318362983713_82393134210.pdf.exe |
| identified language | unknown                                     |
| extracted strings  |                                              |
|   static strings   | 791 (146444 characters)                    |
|   language strings | 0 ( 0 characters)                         |
|   stack strings    | 2                                           |
|   tight strings    | 0                                           |
|   decoded strings  | 66                                          |
+-----+

FLOSS STATIC STRINGS (791)

+-----+
| FLOSS STATIC STRINGS: ASCII (790) |
+-----+
```

remnux@remnux: ~/theZoo/malw... VirusTotal - File - 69e966e730557... 1/2

Right Ctrl



4. Capa

Result:

REMinux v7 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Terminal Aug 1 08:52

remnux@remnux: ~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013

remnux@remnux:~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013\$ capa invoice_2318362983713_8239313421o.pdf.exe
main.py:1109: DeprecationWarning: This is the last capa version supporting Python 3.8 and 3.9.

md5	ea039a054d20d7734c5add48f1a51c34
sha1	9615dca4c0e46b8a39de5428af7db060399230b2
sha256	69e966e730557fde8fd84317cdef1ece00a8bb3470c0b58f3231e170168af169
analysis	static
os	windows
format	pe
arch	1386
path	/home/remnux/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013/invoice_2318362983713_8239313421o.pdf.exe

ATT&CK Tactic	ATT&CK Technique
DEFENSE EVASION	Virtualization/Sandbox Evasion::System Checks [T1497.001]

MBC Objective	MBC Behavior
ANTI-BEHAVIORAL ANALYSIS	Virtual Machine Detection [B0009]

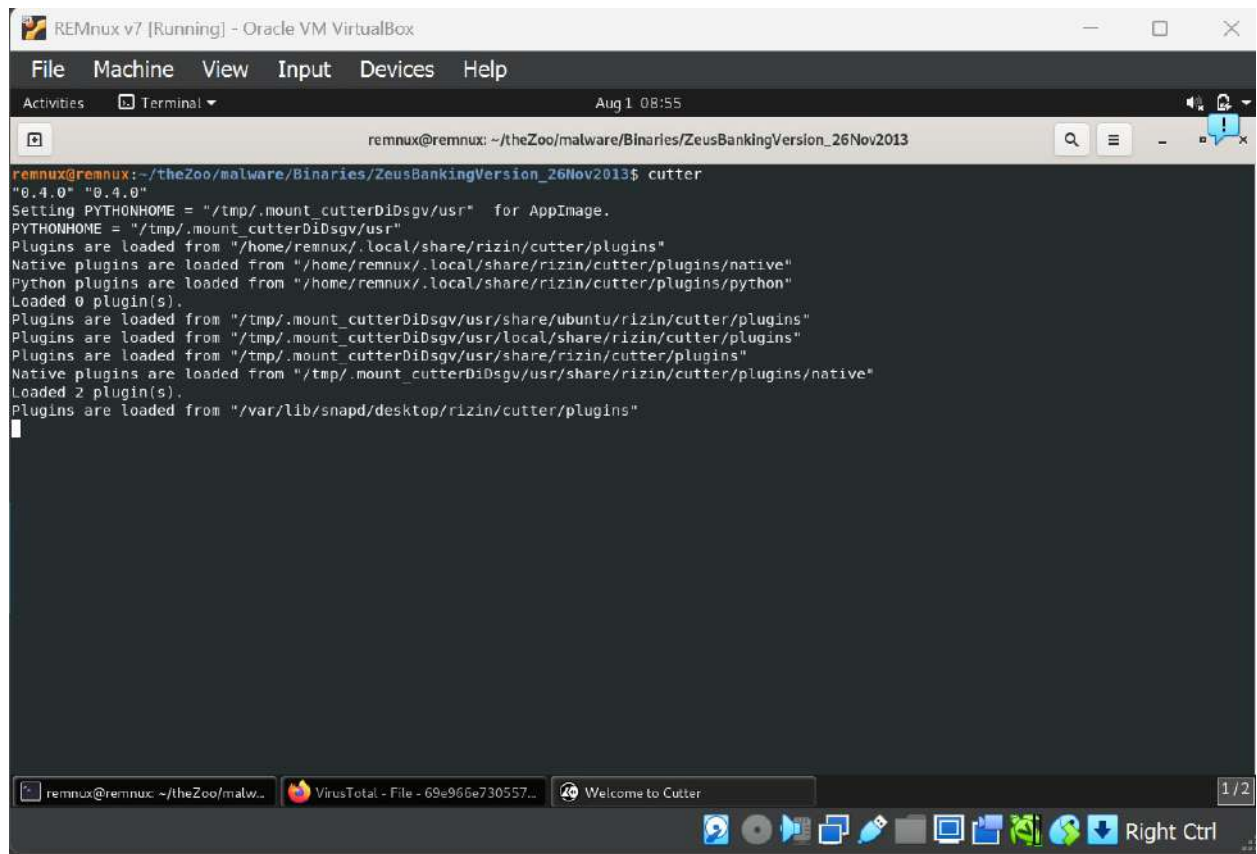
Capability	Namespace
reference anti-vm strings targeting vmware resolve function by parsing PE exports (2 matches)	anti-analysis/anti-vm/vm-detection load-code/pe

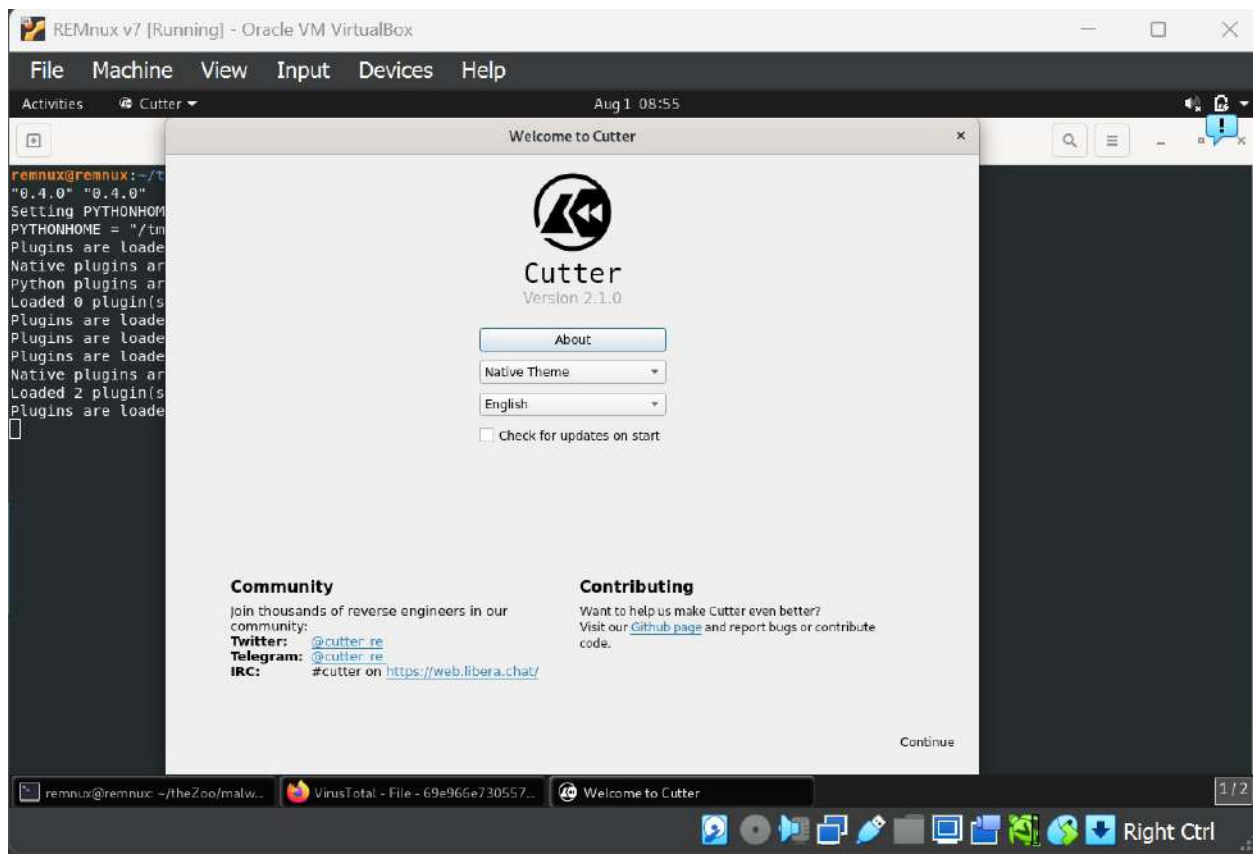
remnux@remnux:~/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013\$

remnux@remnux: ~/theZoo/malw... VirusTotal - File - 69e966e730557... 1/2

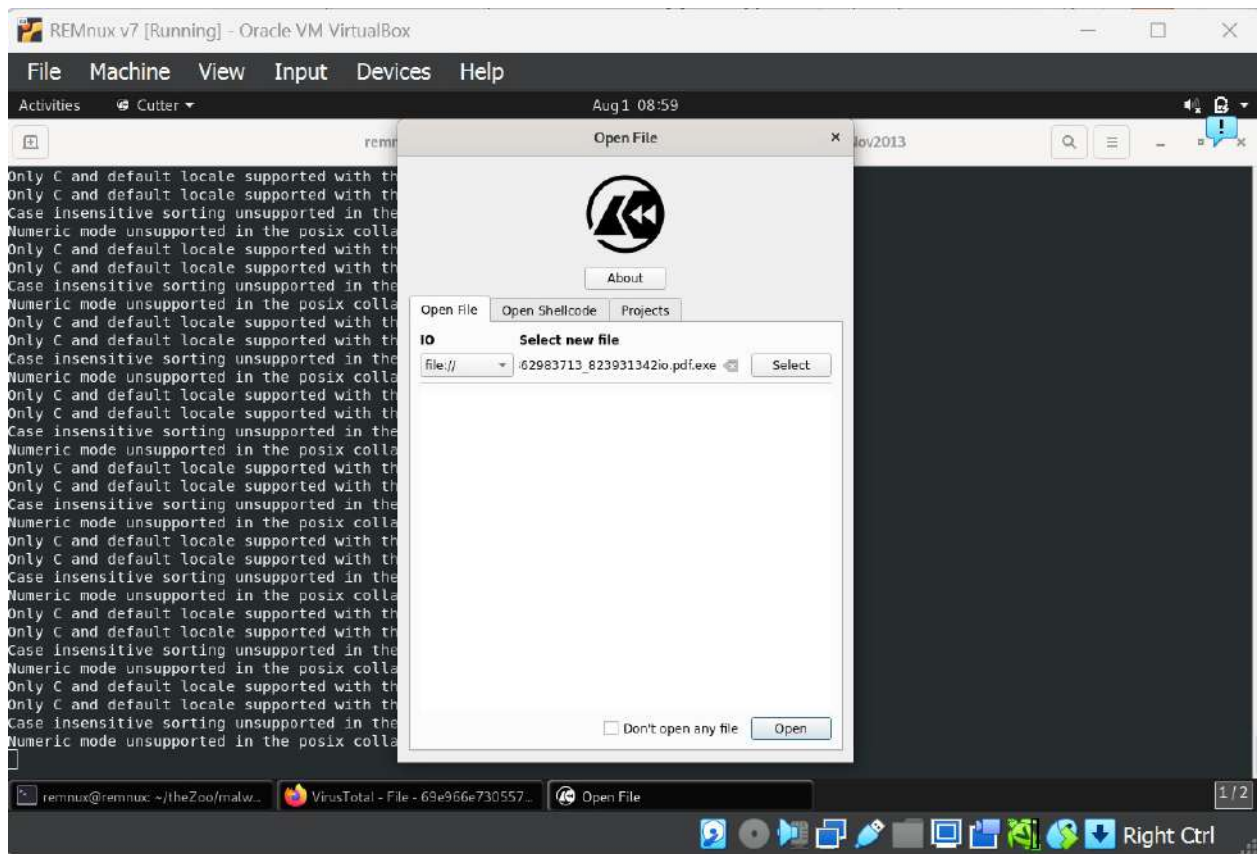
Right Ctrl

5. Cutter





Selected the required file for malware analysis



Cutter Result:

REMnux v7 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Activities Aug 1 09:02

Cutter - /home/remnux/theZoo/malware/Binaries/ZeusBankingVersion_26Nov2013/invoice_2318362983713_823931342io.pdf.exe

File Edit View Windows Debug Help

Type flag name or address here

Functions Dashboard

Name

- entry0
- fcn.004012df
- fcn.0040142f
- fcn.0040166b
- fcn.00401a5f
- fcn.00402aaf
- fcn.0040592c
- fcn.00407123
- fcn.00407436
- fcn.00408469
- fcn.004094b1
- fcn.0040b781
- fcn.0040b975
- fcn.0040b9b0
- fcn.0040b9be
- fcn.0040ba1a
- fcn.0040baf9
- fcn.0040bb32
- fcn.0040bb92
- fcn.0040bba4
- fcn.0040bbff
- fcn.0040bc18
- fcn.0040bc39
- fcn.0040bce3

Quick Filter X

OVERVIEW

Info

File:	/home/remnux/theZoo/malw	FD:	3	Architecture:	x86
Format:	pe	Base addr:	0x00400000	Machine:	i386
Bits:	32	Virtual addr:	True	OS:	windows
Class:	PE32	Canary:	False	Subsystem:	Windows GUI
Mode:	r-x	Crypto:	False	Stripped:	False
Size:	247 kB	NX bit:	False	Relocs:	False
Type:	EXEC (Executable file)	PIC:	False	Endianness:	LE
Language:	c	Static:	False	Compiled:	Mon Nov 25 05:32:03 201
		Relro:	N/A	Compiler:	N/A

Certificates Version info

Hashes

MD5:	ea039a854d20d7734c5add48f1a51c34
SHA1:	9615dca4c0e46b8a39de5428af7db060399230b2
SHA256:	69e966e730557ide8fd84317cdef1ece00a8bb3470c0b58f3231e1701
CRC32:	b6012d5e
ENTROPY:	6.981984

Libraries

- shlwapi.dll
- kernel32.dll
- user32.dll

Dashboard Strings Imports Search Disassembly Graph (entry0) Hexdump Decompiler (entry0)

remnux@remnux: ~/theZoo/malw... VirusTotal - File - 69e966e730557... Cutter - /home/remnux/theZoo/m... 1/2

Right Ctrl

fcn.0040b9b0

fcn.0040b9be

fcn.0040ba1a

fcn.0040baf9

fcn.0040bb32

fcn.0040bb92

fcn.0040bba4

fcn.0040bbff

fcn.0040bc18

fcn.0040bc39

fcn.0040bce3

Quick Filter X

Analysis info

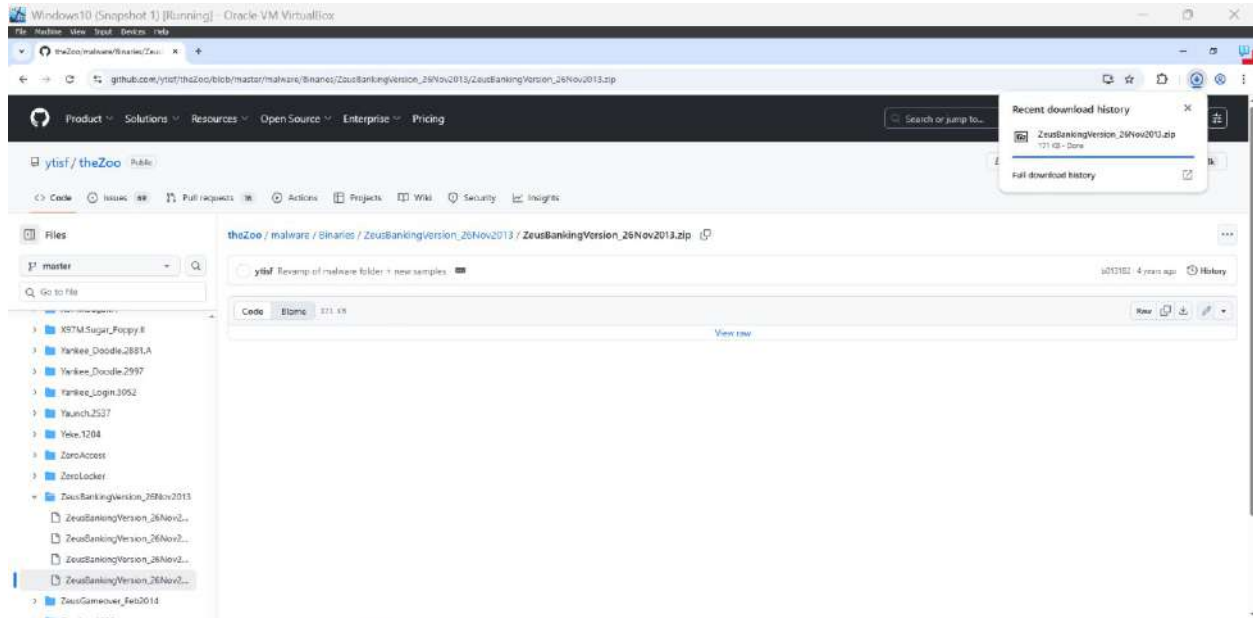
Functions:	64
X-Refs:	2534
Calls:	2406
Strings:	554
Symbols:	77
Imports:	77
Analysis coverage:	44912 bytes
Code size:	150016 bytes
Coverage percent:	29%

Dashboard Strings Imports Search Disassembly Graph (entry0) Hexdump Decompiler (entry0)

remnux@remnux: ~/theZoo/malw... VirusTotal - File - 69e966e730557... Cutter - /home/remnux/theZoo/m... 1/2

Right Ctrl

Downloading the zip file of zeus inside FlareVM

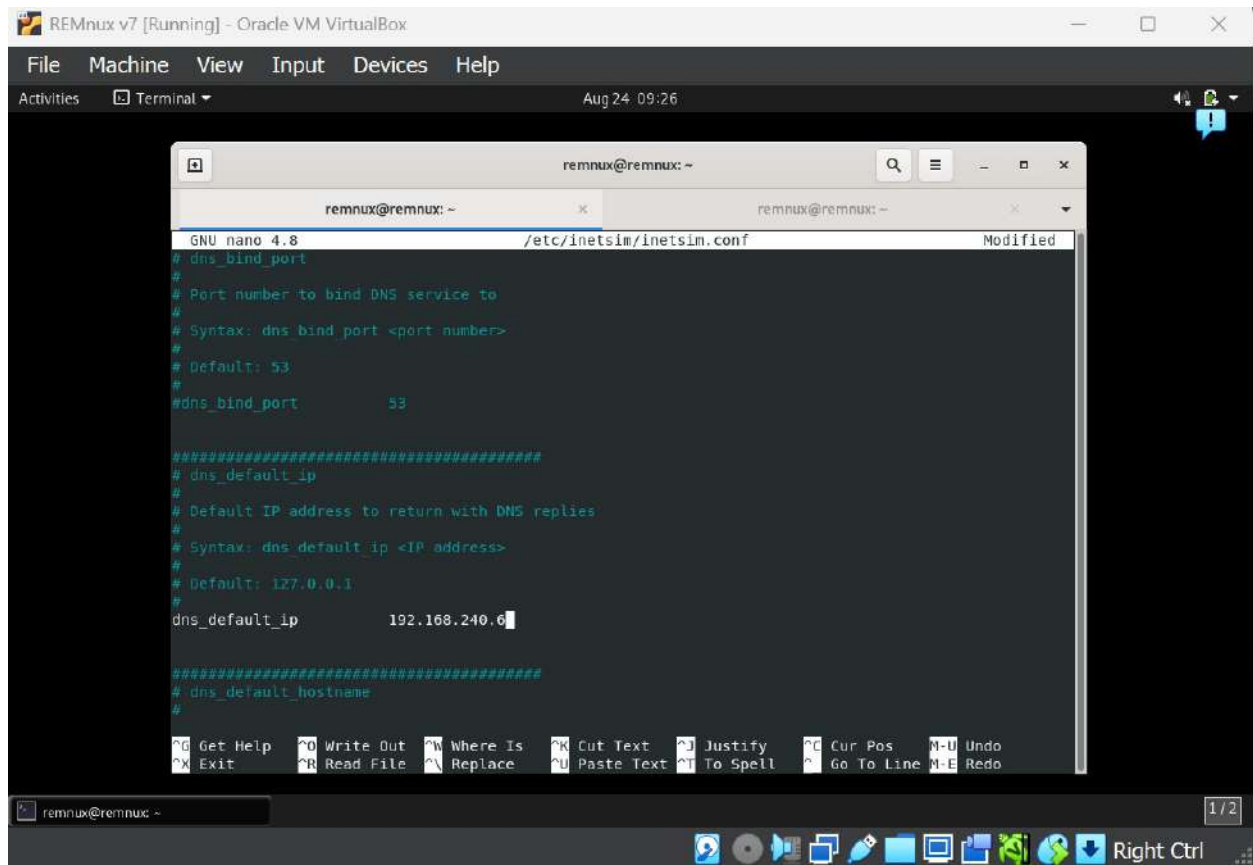


Saved the exe file in the following directory

C:\Users\Rahul\Downloads\Zeus\invoice_231836298371
3_823931342io.pdf.exe

6. INetSim

Remnux ip: 192.168.240.6



```
REMnux v7 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Aug 24, 09:26

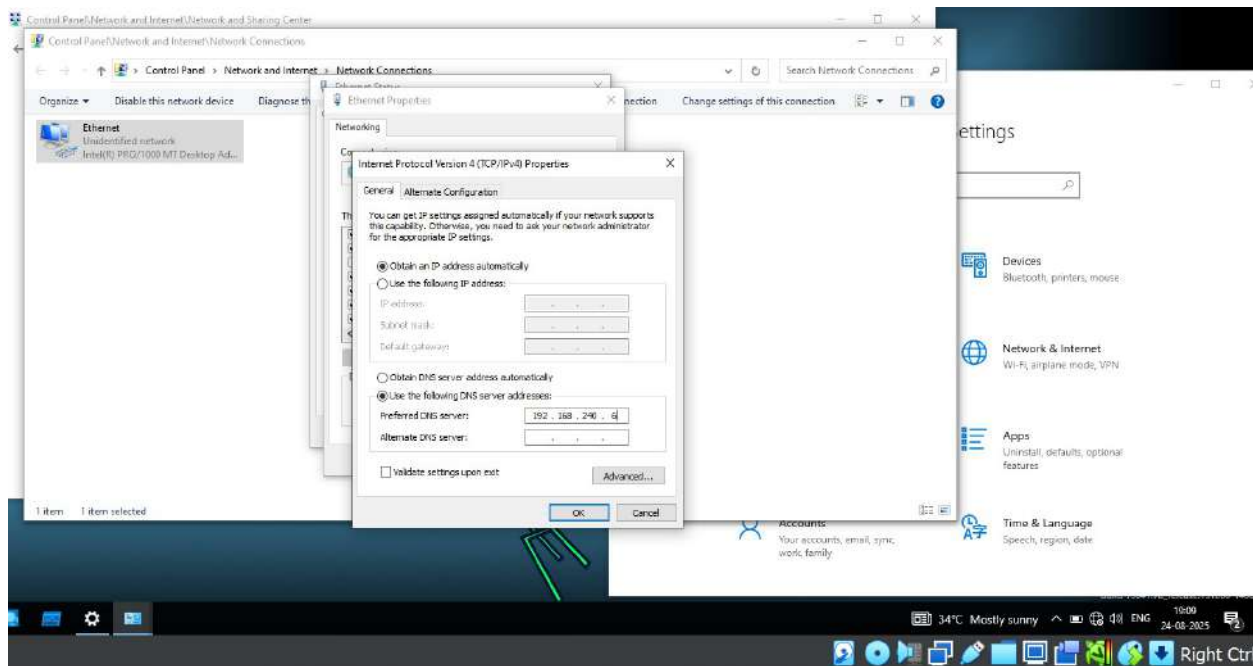
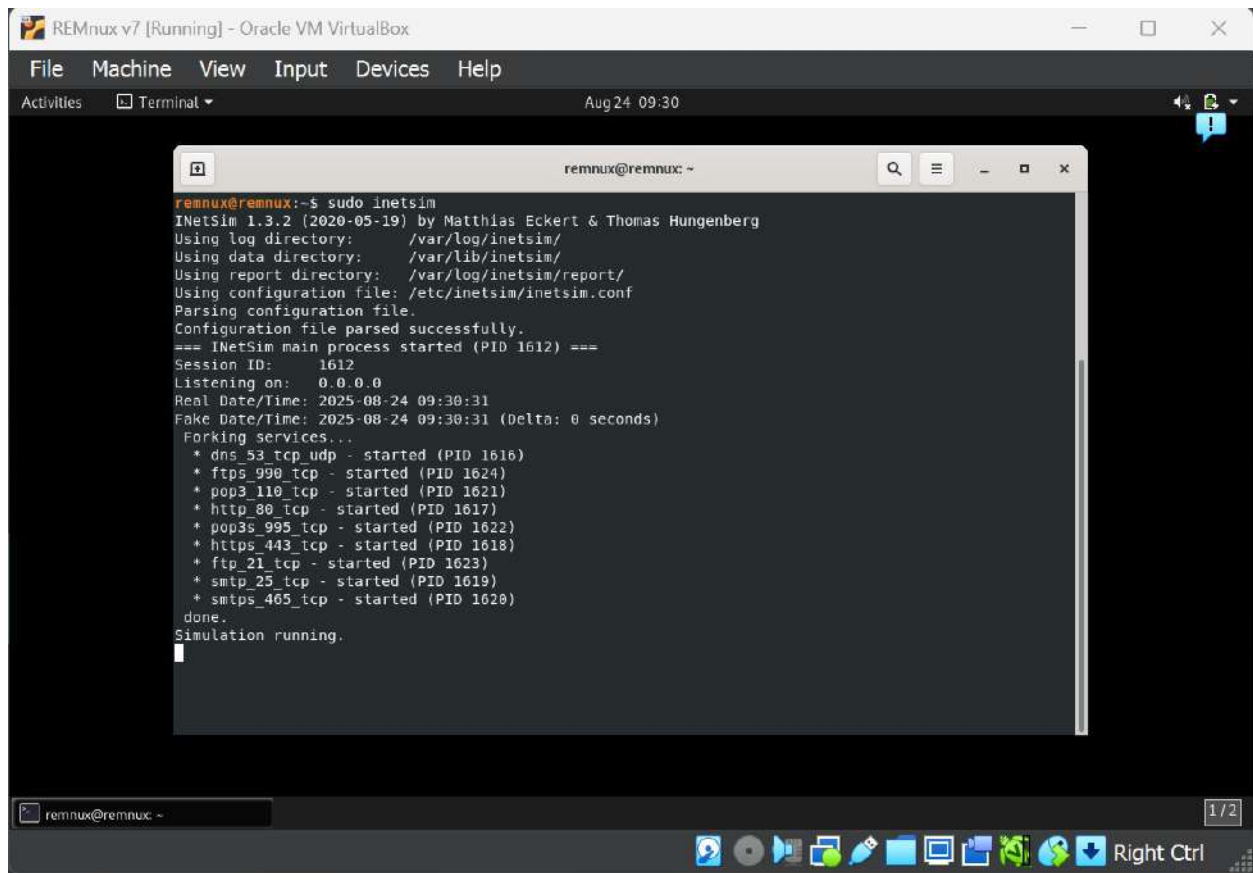
remnux@remnux: ~
GNU nano 4.8 /etc/inetsim/inetsim.conf Modified
# dns_bind_port
#
# Port number to bind DNS service to
#
# Syntax: dns_bind_port <port number>
#
# Default: 53
#
#dns_bind_port      53

#####
# dns_default_ip
#
# Default IP address to return with DNS replies
#
# Syntax: dns_default_ip <IP address>
#
# Default: 127.0.0.1
#
dns_default_ip      192.168.240.6

#####
# dns_default_hostname
#

^G Get Help  ^O Write Out ^W Where Is  ^K Cut Text  ^J Justify   ^C Cur Pos   ^M-U Undo
^X Exit      ^R Read File ^N Replace   ^U Paste Text ^T To Spell  ^G Go To Line ^M-E Redo

remnux@remnux: ~ 1/2
```



```
REMnux v7 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Aug 24 10:00

remnux@remnux: /var/log/inetsim
remnux@remnux: ~
remnux@remnux: /var/log/inetsim

remnux@remnux: /var/log/inetsim$ tail -f service.log
tail: cannot open 'service.log' for reading: Permission denied
tail: no files remaining
remnux@remnux: /var/log/inetsim$ tail -f http.log
tail: cannot open 'http.log' for reading: No such file or directory
tail: no files remaining
remnux@remnux: /var/log/inetsim$ tail -f dns.log
tail: cannot open 'dns.log' for reading: No such file or directory
tail: no files remaining
remnux@remnux: /var/log/inetsim$ sudo tail -f /var/log/inetsim/service.log
[2025-08-24 09:52:59] [1612] [dns_53_tcp_udp 1616] [192.168.240.7] recv: Query Type A, Class IN, Name tsfe.trafficshaping.dsp.mp.microsoft.com
[2025-08-24 09:52:59] [1612] [dns_53_tcp_udp 1616] [192.168.240.7] send: tsfe.trafficshaping.dsp.mp.microsoft.com 3600 IN A 192.168.240.6
[2025-08-24 09:52:59] [1612] [dns_53_tcp_udp 1616] [192.168.240.7] disconnect
[2025-08-24 09:52:59] [1612] [dns_53_tcp_udp 1616] [192.168.240.7] stat: 1 qtype=A qclass=IN qname=tsfe.trafficshaping.dsp.mp.microsoft.com
[2025-08-24 09:52:59] [1612] [https_443_tcp 1704] [192.168.240.7:49705] connect
[2025-08-24 09:52:59] [1612] [https_443_tcp 1704] [192.168.240.7:49705] info: Error setting up SSL: SSL accept attempt failed
[2025-08-24 09:52:59] [1612] [https_443_tcp 1704] [192.168.240.7:49705] disconnect
[2025-08-24 09:55:55] [1612] [https_443_tcp 1710] [192.168.240.7:49706] connect
[2025-08-24 09:55:55] [1612] [https_443_tcp 1710] [192.168.240.7:49706] info: Error setting up SSL: SSL accept attempt failed
[2025-08-24 09:55:55] [1612] [https_443_tcp 1710] [192.168.240.7:49706] disconnect
[2025-08-24 09:59:33] [1612] [https_443_tcp 1717] [192.168.240.7:49707] connect
[2025-08-24 09:59:33] [1612] [https_443_tcp 1717] [192.168.240.7:49707] info: Error setting up SSL: SSL accept attempt failed
[2025-08-24 09:59:33] [1612] [https_443_tcp 1717] [192.168.240.7:49707] disconnect
```

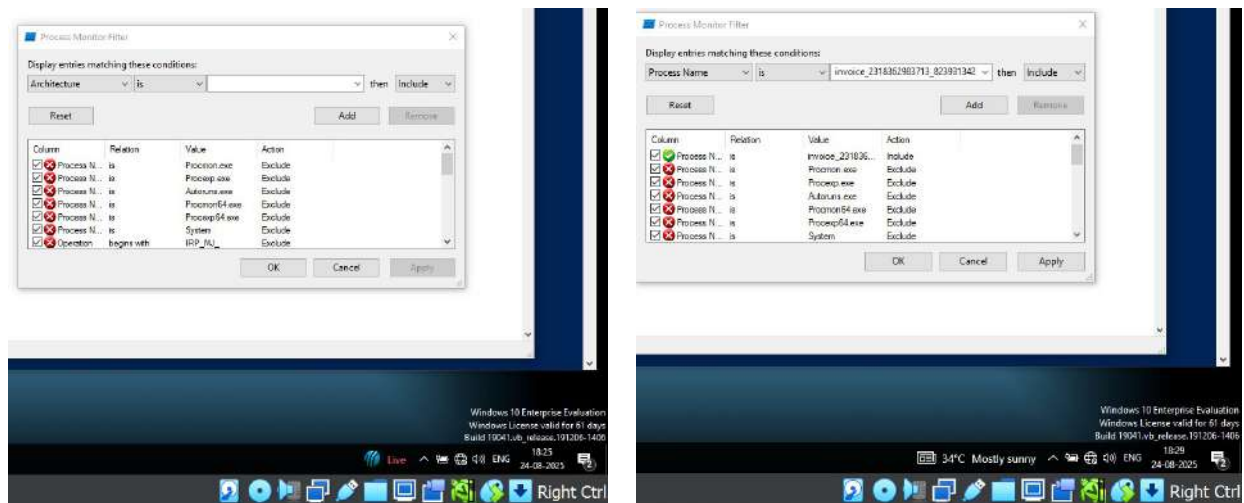
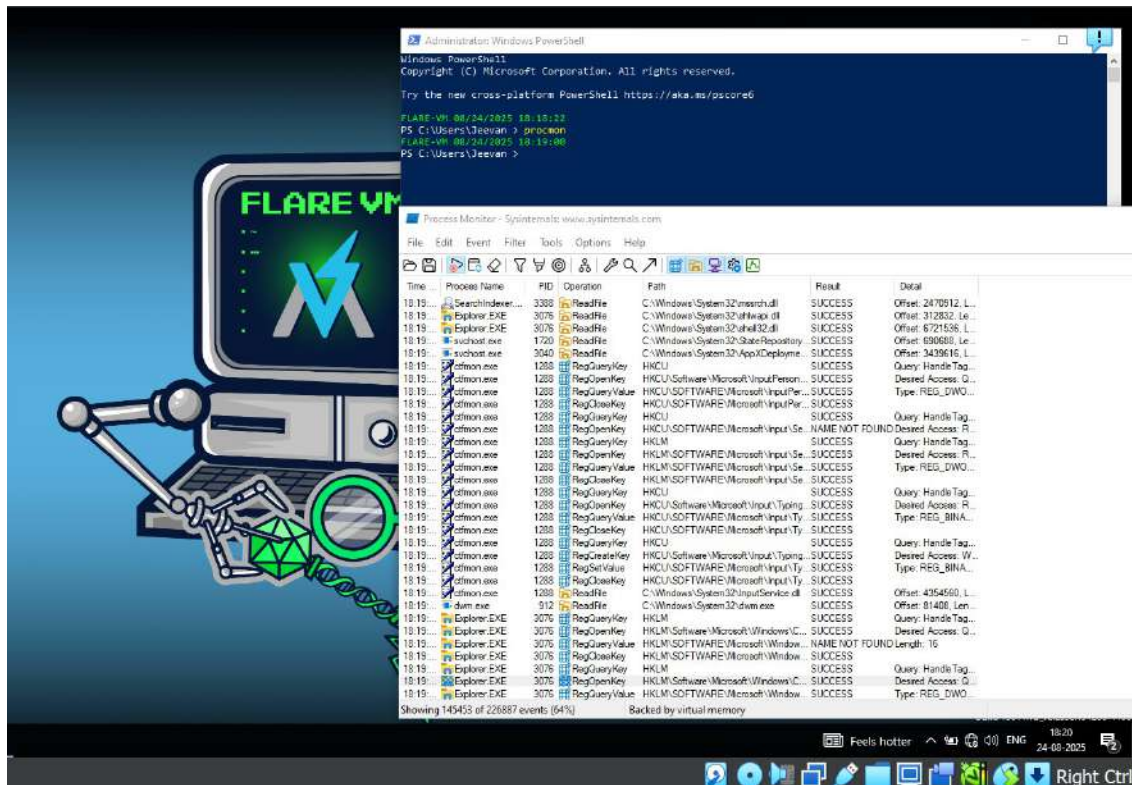
InetSim Result:

```
REMnux v7 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities Terminal Aug 24 10:02

remnux@remnux: /var/log/inetsim
remnux@remnux: ~
remnux@remnux: /var/log/inetsim

remnux@remnux: /var/log/inetsim$ sudo tail -f /var/log/inetsim/service.log
[2025-08-24 09:52:59] [1612] [dns_53_tcp_udp 1616] [192.168.240.7] stat: 1 qtype=A qclass=IN qname=tsfe.trafficshaping.dsp.mp.microsoft.com
[2025-08-24 09:52:59] [1612] [https_443_tcp 1704] [192.168.240.7:49705] connect
[2025-08-24 09:52:59] [1612] [https_443_tcp 1704] [192.168.240.7:49705] info: Error setting up SSL: SSL accept attempt failed
[2025-08-24 09:52:59] [1612] [https_443_tcp 1704] [192.168.240.7:49705] disconnect
[2025-08-24 09:55:55] [1612] [https_443_tcp 1710] [192.168.240.7:49706] connect
[2025-08-24 09:55:55] [1612] [https_443_tcp 1710] [192.168.240.7:49706] info: Error setting up SSL: SSL accept attempt failed
[2025-08-24 09:55:55] [1612] [https_443_tcp 1710] [192.168.240.7:49706] disconnect
[2025-08-24 09:59:33] [1612] [https_443_tcp 1717] [192.168.240.7:49707] connect
[2025-08-24 09:59:33] [1612] [https_443_tcp 1717] [192.168.240.7:49707] info: Error setting up SSL: SSL accept attempt failed
[2025-08-24 09:59:33] [1612] [https_443_tcp 1717] [192.168.240.7:49707] disconnect
```

7.ProcMon



Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time	Process Name	PID	Operation	Path	Result	Detail
18:29...	Invoice_23183...	2408	Process Start		SUCCESS	Parent PID: 3076, ...
18:29...	Invoice_23183...	2408	Thread Create		SUCCESS	Thread ID: 1700
18:29...	Invoice_23183...	2408	Load Image	C:\Users\Jeevan\Downloads\Zeus\Inv...	SUCCESS	Image Base: 0x400...
18:29...	Invoice_23183...	2408	Load Image	C:\Windows\System32\ntdll.dll	SUCCESS	Image Base: 0x7fb...
18:29...	Invoice_23183...	2408	Load Image	C:\Windows\System32\wow64\ntdll.dll	SUCCESS	Image Base: 0x770...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 00
18:29...	Invoice_23183...	2408	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Con...	REPARSE	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 24
18:29...	Invoice_23183...	2408	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	
18:29...	Invoice_23183...	2408	CreateFile	C:\Windows	SUCCESS	Desired Access: E...
18:29...	Invoice_23183...	2408	Load Image	C:\Windows\System32\wow64.dll	SUCCESS	Image Base: 0x7fb...
18:29...	Invoice_23183...	2408	Load Image	C:\Windows\System32\wow64\win.dll	SUCCESS	Image Base: 0x7fb...
18:29...	Invoice_23183...	2408	QueryOpen	C:\Windows\System32\wow64\log.dll	NAME NOT FOUND	
18:29...	Invoice_23183...	2408	CreateFile	C:\Windows	SUCCESS	Desired Access: R...
18:29...	Invoice_23183...	2408	QueryNameInfo	C:\Windows	SUCCESS	Name: \Windows
18:29...	Invoice_23183...	2408	CloseFile	C:\Windows	SUCCESS	
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\Software\Microsoft\Wow64\B6	SUCCESS	Desired Access: R...
18:29...	Invoice_23183...	2408	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	NAME NOT FOUND	Length: 520
18:29...	Invoice_23183...	2408	RegQueryValue	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	Type: REG_SZ, Lo...
18:29...	Invoice_23183...	2408	RegCloseKey	HKLM\SOFTWARE\Microsoft\Wow64\...	SUCCESS	
18:29...	Invoice_23183...	2408	Load Image	C:\Windows\System32\wow64\cpu.dll	SUCCESS	Image Base: 0x770...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	REPARSE	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegOpenKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	Desired Access: Q...
18:29...	Invoice_23183...	2408	RegSetInfoKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	KeySelfInformation...
18:29...	Invoice_23183...	2408	RegQueryValue	HKLM\System\CurrentControlSet\Contr...	NAME NOT FOUND	Length: 00
18:29...	Invoice_23183...	2408	RegCloseKey	HKLM\System\CurrentControlSet\Contr...	SUCCESS	

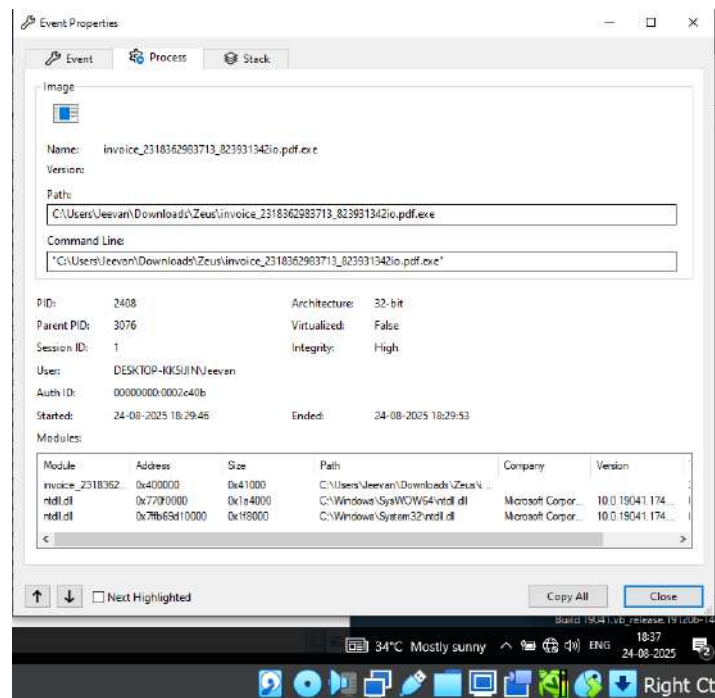
Showing 869 of 173458 events (0.50%) Backed by virtual memory

Build: 19041.vb_release.191205-1408

34°C Mostly sunny 18:31 24-03-2025 ENG

Right Ctrl

- The process invoice_2318362983713_823931342io.pdf.exe started correctly (Process Start).
- It loaded system DLLs (wow64.dll, wow64cpu.dll, kernel32.dll, etc.) → typical of an EXE initializing.
- It touched registry keys under:
 - HKLM\System\CurrentControlSet\Control\...
 - HKLM\Software\Microsoft\Wow64...
- It created files (CreateFile, CreateFileMapping) where CreateFile especially shows the path of dropping a copy



8) Wireshark

1. VirusTotal

- File scanned with multiple antivirus engines. Detected as Zeus variant (Trojan/Spyware). Shows hash values (MD5, SHA1, SHA256) and network IOCs (malicious URLs, IPs).

2. PeStudio

- Suspicious imports like CreateProcess, WriteFile, InternetOpenUrl. Indicators of obfuscation and packer usage. Shows embedded resources and potential C2 communication endpoints.

3. FLOSS

- Extracted hidden strings include banking URLs, login forms, and system commands. Revealed hardcoded IP addresses and domain names for exfiltration.

4. Capa

- Identified capabilities: keylogging, process injection, HTTP/HTTPS communication, persistence mechanisms. Confirms malware attempts to evade detection and gather sensitive user data.

5. Cutter

- Reverse-engineering confirms routines for credential stealing. Shows functions for hooking system APIs, network communication, and encrypted payload handling.

6. INetSim

- Malware attempted to connect to simulated C2 servers. Confirms network behavior and command requests over HTTP/HTTPS.

7. Process Monitor

- File system and registry modifications detected. Creates startup entries and injects into running processes. Logs reveal creation of temporary files for execution.

8. Wireshark

- Captured packets show encrypted communications to remote servers. Unusual traffic patterns detected, consistent with data exfiltration attempts.

Inferences: Zeus Banking Trojan Analysis

Zeus is a banking Trojan that primarily targets financial information. Creates registry entries to run on startup. Copies itself to system directories to maintain persistence. Hooks into Windows processes to ensure it continues running after reboot. It is a sophisticated malware targeting banking credentials through keylogging, form grabbing, and browser injection.

It uses persistence mechanisms (registry changes, process injection) to remain on the system. Malware communicates with C2 servers over encrypted channels to exfiltrate sensitive data. Analysis with multiple tools confirms its capabilities: credential theft, network communication, process injection, and evasion techniques. Timely detection, network monitoring, and updated antivirus signatures are critical to mitigate the threat posed by Zeus.

Q2.

```
rule Zeus_Banking_Trojan
{
    meta:
```

```
description = "Detects Zeus Banking Trojan (Zbot)"
author = "Jeevan"
date = "2025-08-24"
reference =
"https://github.com/ytisf/theZoo/tree/master/malware/Binaries/ZeusBankingVersion_26Nov2013"
```

strings:

```
$s1 = "Zeus" ascii wide
$s2 = "sdra64.exe" ascii wide
$s3 = "Botnet" ascii nocase
$s4 = "User-Agent" ascii wide
$s5 = "config.bin" ascii
$s6 = "Software\\Microsoft\\Windows\\CurrentVersion\\Run" wide
```

```
$api1 = "CreateProcessW" ascii
$api2 = "WinExec" ascii
$api3 = "InternetOpenA" ascii
$api4 = "InternetConnectA" ascii
$api5 = "HttpSendRequestA" ascii
```

condition:

```
uint16(0) == 0x5A4D and
( 3 of ($s*) or 2 of ($api*) )
}
```